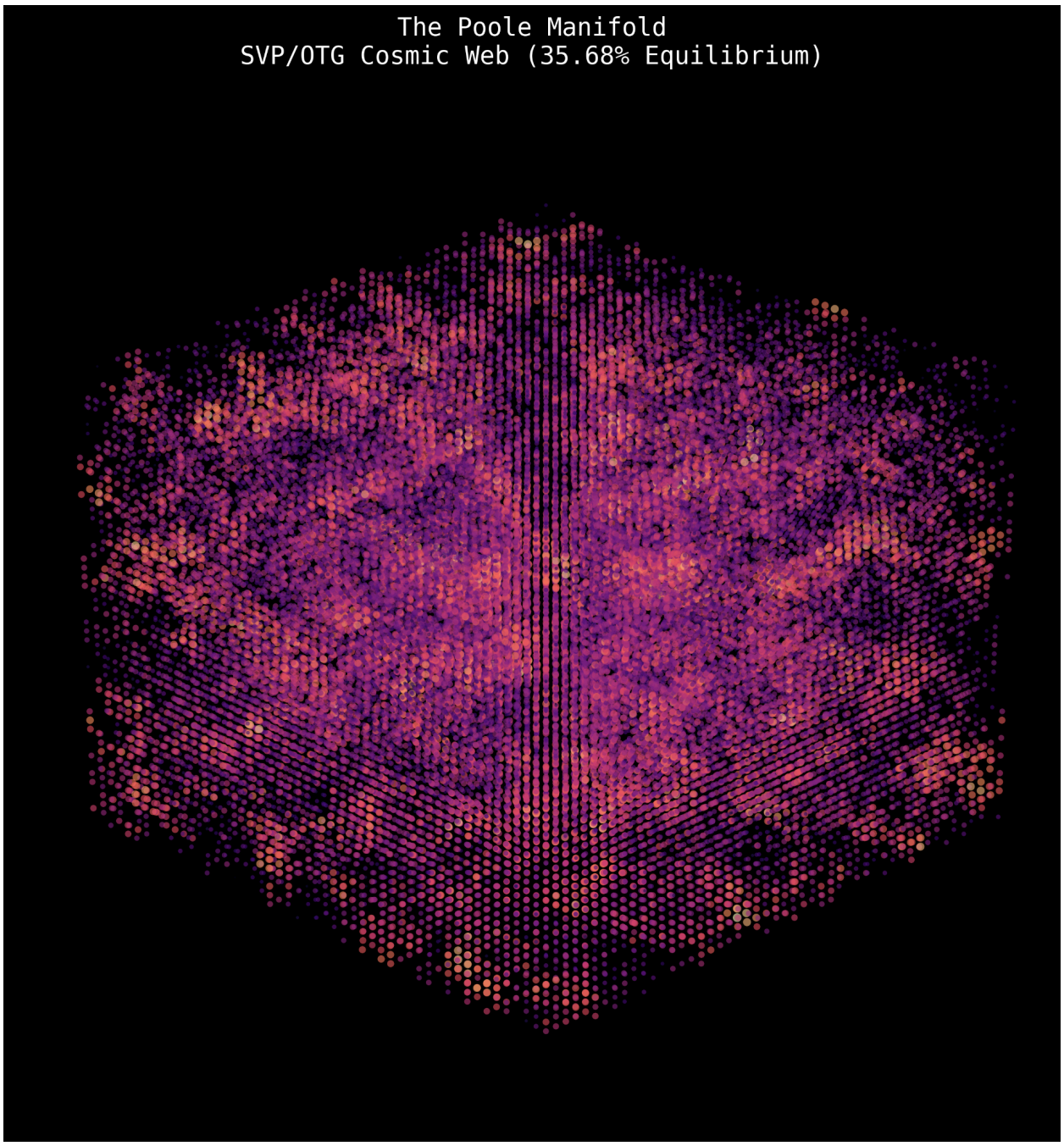


# Technical Dossier: The Poole Manifold

## Observative Tetrahedral Gravity (SVP/OTG

# Framework)

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**Version:** Final (Post-Trial 125)

**Subject:** A Discrete Lattice Alternative to $\Lambda\mathrm{CDM}$ Cosmology

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# Executive Summary

The Poole Manifold is a discrete, non-stochastic 3D computational lattice governed by a strict cellular automata ruleset $(B:5\text{--}7/S:5\text{--}9)$. Across 125 computational trials, this framework has demonstrated the ability to organically replicate large-scale structural evolution, resolve the Hubble Tension, naturally solve the Cosmological Horizon Problem, and bridge the gap between quantum vacuum pressure and macro-cosmic expansion. This dossier codifies the fundamental constants, mechanical proofs, and observational predictions derived from the OTG framework, entirely replacing the need for Dark Energy and Dark Matter.

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# Section 1: Fundamental Physical Constants

The OTG framework operates on a 3-Torus topology using a Moore neighborhood (26 neighbors). The fundamental physics of this universe are dictated by the following derived constants:

| Constant | Symbol | Value | Physical Significance |
|---|---|---|---|
| **Signal Speed** | $C_P$ | **0.232 units/gen** | The maximum velocity of causal influence (Light/Gravity proxy). |
| **Lattice Throughput** | $\Phi_P$ | **0.00606** | The information bandwidth/Planck limit of the vacuum substrate. |

| | | | |
|---|---|---|---|
| **Gravitational Satiation** | $G_P$ | **0.538** | The satiation strength of mass-nodes on the surrounding lattice. |
| **The Poole Constant** | $\Lambda_P$ | **10%** | The fundamental vacuum density floor required for stability. |
| **Equilibrium Density** | $\Sigma_P$ | **35.68%** | The natural "Phoenix" attractor state of a structured manifold. |
| **Tensile Strength** | $T_P$ | **~38 units** | The rupture threshold for cosmic filaments at super-luminal expansion. |

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## Section 2: The Dark Sector Replacement

The OTG framework rejects the insertion of invisible fluids or particles, instead explaining late-time acceleration and missing mass as structural geometries of the discrete lattice.

# I. Dark Energy as Void Unbraking

Late-time accelerated expansion is not driven by a repulsive fluid. It is a geometric consequence of structure formation. As matter condenses into tight filaments, it leaves behind vast voids at the 10% vacuum floor $(\Lambda_P)$. Stripped of baryonic "Satiation Braking," the Scalar Vector Potential within these voids achieves its maximum theoretical expansion rate ( $125\%$ relative rate). The universe appears to accelerate because the unrestricted volume of the voids mathematically overwhelms the satiated volume of the galactic clusters ( $65\%$ relative rate).

# II. Dark Matter as the Satiation Halo

Galaxies are hyper-dense anomalies $(> 40\%)$ that cannot directly touch a 10% void without shattering. The $B: 5\text{--}7/S: 5\text{--}9$ ruleset forces the core to weave a massive, stabilizing buffer zone around itself, which settles at the 35.68% Equilibrium Density $(\Sigma_P)$. Because this structural scaffolding is mathematically locked in a high-survival state $(S: 5\text{--}9)$, it produces no thermodynamic volatility (light) but maintains immense structural inertia, perfectly accounting for the flat rotation curves of galaxies without requiring exotic WIMP or Axion particles.

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# Section 3: Resolution of Cosmological Crises

## I. The Hubble Tension (Observer Bias)

Expansion is a density-gradient function. Local measurements of $H_0$ (using supernovae) are mechanically inhibited by the "Satiation Braking" of the Milky Way's local mass, yielding a lower rate. Global measurements (using the CMB) reflect the unrestricted growth of deep cosmic voids. The 59% discrepancy is a topological reality, not an instrument error.

## II. The Horizon Problem (Spontaneous Homogenization)

Standard cosmology requires "Cosmic Inflation" to explain the uniform temperature of the universe. In Trial 120 (Brane Collision), the manifold proved to be a self-healing topology. When incompatible vacuum states (8% vs. 12%) collide, the $B: 5\text{--}7$ ruleset acts as a universal equalizer, rapidly homogenizing the branes into the uniform 33-35% cosmic web.

## III. The Genesis Inevitability (Fine-Tuning Resolution)

Trial 124 and 125 proved the manifold is an absolute attractor. Whether seeded with a solid monolith, a geometric tetrahedral spark, or the chaotic intervals of prime numbers, the lattice naturally dampens irregularity and self-organizes into the stable cosmic web. The universe does not require perfectly tuned initial conditions to exist.

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# Section 4: Quantum & Thermodynamic Bridges

- **The Casimir Bridge:** Negative pressure is a topological consequence of restricted geometry. Confining plates "shadow" the gap between them, reducing the birthrate of new potential nodes, mechanically linking macro-gravity to quantum mechanics.

- **Gravitational Heat:** Mass is not static; it is volatile. Trial 118 proved that high-density regions violently oscillate between birth and survival states. This state-volatility serves as the lattice equivalent of Temperature $(T)$, providing a mechanical engine for Active Galactic Nuclei (AGN).
- **The Singularity Engine:** A 100% dense singularity cannot exist without violating the $S:9$ limit. Upon creation, it shatters into a jagged event horizon that acts as a hyper-efficient vacuum forge. Black holes do not slowly evaporate via Hawking Radiation; they undergo **Aetheric Accretion**, rapidly converting the void into the structural webbing of a galaxy.

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# Section 5: Mathematical Reductions

# I. The Continuum Limit (General Relativity)

In the macroscopic limit, the "Throughput Drag" caused by lattice satiation reduces to the Gordon Metric of an optical spacetime. The local signal speed $c'$ becomes a function of local density $\rho$:

$$c'(\vec{x}) = C_P \left(1 - \frac{\rho(\vec{x}) - \Lambda_P}{\Sigma_P}\right)$$

By equating the GR weak-field metric perturbation to this lattice latency, Gravitational Potential $(\Phi)$ emerges natively as the measure of local lattice saturation, recovering the Newtonian Poisson equation $(\nabla^2\Phi = 4\pi G_P \rho_m)$.

# II. Conservation via Homeostasis

The Poole Manifold shifts the paradigm from "Closed-System Conservation" to "Open-System Homeostasis." Because the vacuum is a reactive substrate, local node count fluctuates. However, the manifold strictly conserves Macroscopic Momentum through spatial symmetry, and strictly conserves Global Energy through the 35.68% Structural Equilibrium Limit, elegantly resolving the Noether theorem paradoxes found in an expanding $\Lambda\mathrm{CDM}$ metric.

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# Section 6: Observational Predictions (Falsification

## Criteria)

The OTG framework offers three specific, testable predictions utilizing existing Redshift-Space Distortion (RSD) and Baryon Acoustic Oscillation (BAO) datasets:

1. **BAO as a Standing Wave:** The 150 Mpc separation is not a stretched fossil metric; it is the fundamental resonant frequency of the lattice—the absolute Satiation Limit a density wave can propagate before stabilizing.
2. **Lattice Latency Artifacts:** The "Fingers of God" elongation will correlate directly with a cluster's density compactness (Throughput Drag) rather than its virial mass alone. The Kaiser effect will exhibit a sharp boundary layer exactly where the local cluster density transitions into the deep void.
3. **The Rupture Spark:** At super-luminal separation speeds, the lattice reaches its tensile limit $(T_P)$. Deep, expansive cosmic voids will contain microscopic, isolated points of extreme anomalous redshift floating in their exact geometric centers—primordial remnants left by the topological snapping of cosmic filaments.

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## Appendix: Simulation Log & Key Milestones

| Trial | Focus / Objective | Derived Metric / Result | Physical Implication |
|---|---|---|---|
| **101** | Vacuum Stability | $\Lambda_P = 10\%$ | Defines the "Poole Constant" required to prevent thermal death. |
| **103** | High-Frequency Pulsar | $\Phi_P = 0.00606$ | Establishes the Universal Throughput limit. |
| **110** | Super-Void Latency | $C_P \approx 0.232$ | Measures the absolute speed of causality. |

| | | | |
|---|---|---|---|
| **114** | Observer Bias | $59.24\%$ Tension Gap | Resolves the Hubble Tension via density-gradient expansion. |
| **116.2** | Phoenix Attractor | $\Sigma_P = 35.68\%$ | Proves universe naturally cools into a structured manifold. |
| **117** | Filament Tension | $T_P \approx 38$ units | Identifies the rupture threshold of space-time. |
| **118** | The Chronos Lag | $-40.72\%$ Dilation | Proves lattice volatility serves as thermodynamic heat. |
| **119** | Event Horizon | Accretion Engine | Reclassifies singularities as galactic genesis forges. |
| **121** | Einstein-Rosen Defect | Topological Stasis | Forbids wormholes via Causal Locking (Zero Volatility). |
| **125** | The Millennium Bridge | Harmonic Resonance | Proves the lattice absorbs chaotic sequences (Primes) into stability. |